

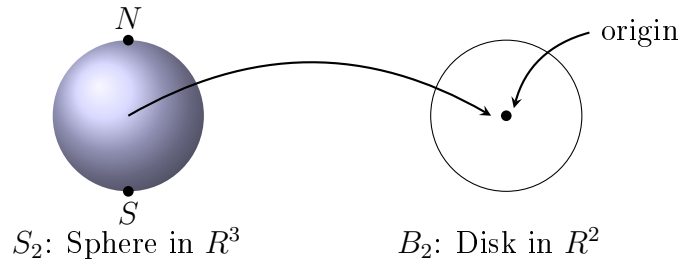
Homework 8: Solutions

Problem 1.

We showed in class that S_3 and B_3 are topologically equivalent (with all points of the boundary of B_3 identified as only point), namely there is a 1-1 map between them which is continuous.

a) Now analyse the case S_2 and B_2 .

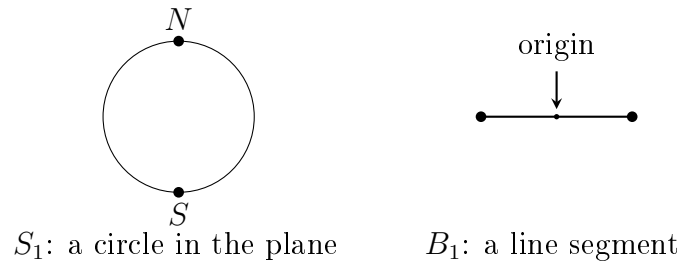
Solution:



We wrap the disk around the sphere: origin of disk at North pole of sphere, the whole boundary of the disk becomes the South pole. So all points of the circle that is the boundary of B_2 are identified.

b) Next analyse the case S_1 and B_1 .

Solution:



Wrap the line segment around the circle. The two endpoints of the line segment must be identified because they become the South pole.

c) In which of these 3 cases is the sphere isomorphic to a group, and which group?

Solution: S_3 is isomorphic to $SU(2)$. S_2 is not a group (the LA with two generators is abelian, namely two commuting generators A and B). S_1 is the “circle group”, namely $U(1)$.

d) Which of these groups is a covering group, and which of them is not simply-connected?

Solution: $SU(2)$ is a covering group (of $SO(3)$) and is simply-connected. S_1 is not simply connected, so it is **not** a covering group. S_1 and B_1 are topologically the same, their covering group is R^1 (unwind the circle, as discussed in class).

Problem 2.

We proved that $SU(2)/Z_2 \simeq SO(3)$ by considering the equation

$$u^{-1}x^k\sigma_k u = x'^k\sigma_k = R^i_j x^j\sigma_i, \quad u = e^{-\frac{i}{2}\omega^k\sigma_k}. \quad (1)$$

- a) Prove that the left-hand side is linear in σ_l (expand $e^{-\frac{i}{2}\omega^k\sigma_k}$) and determine R^i_j .

Solution: We have to show that expanding u we only get terms with σ_k but no terms with I .

$$\begin{aligned} \left(\cos \frac{\omega}{2} + i \frac{\omega^l}{\omega} \sin \frac{\omega}{2} \sigma_l \right) \sigma_k \left(\cos \frac{\omega}{2} - i \frac{\omega^m}{\omega} \sin \frac{\omega}{2} \sigma_m \right) \\ = \cos^2 \frac{\omega}{2} \sigma_k + \left[i \frac{\omega^l}{\omega} \sin \frac{\omega}{2} \sigma_l, \sigma_k \right] \cos \frac{\omega}{2} + \frac{\omega^l}{\omega} \frac{\omega^m}{\omega} \sin^2 \frac{\omega}{2} \sigma_l \sigma_k \sigma_m \end{aligned} \quad (2)$$

Using $\omega^l \omega^m \sigma_l \sigma_k \sigma_m = \omega_k \omega^m \sigma_m + \omega^l \sigma_l \omega_k - \omega^2 \sigma_k$ we get

$$\begin{aligned} &= \left(\cos^2 \frac{\omega}{2} - \sin^2 \frac{\omega}{2} \right) \sigma_k + i \frac{\omega^l}{\omega} \sin \frac{\omega}{2} \cos \frac{\omega}{2} (2i \epsilon_{lkm} \sigma_m) + 2 \frac{\omega^k \omega^l}{\omega^2} \sigma_l \sin^2 \frac{\omega}{2} \\ &= \cos \omega \sigma_k + \epsilon_{klm} \frac{\omega^l}{\omega} \sigma_m \sin \omega + \frac{\omega^k \omega^l}{\omega^2} \sigma_l (1 - \cos \omega) \\ &= \left(\delta^{kl} - \frac{\omega^k \omega^l}{\omega^2} \right) \cos \omega \sigma_l + \frac{\omega^k \omega^l}{\omega^2} \sigma_l + \epsilon_{klm} \frac{\omega^l}{\omega} \sin \omega \sigma^m \equiv R^k_l \sigma^l \\ \Rightarrow R^k_l &= \left(\delta^k_l - \frac{\omega^k \omega^l}{\omega^2} \right) \cos \omega + \frac{\omega^k \omega^l}{\omega^2} - \epsilon_{klm} \frac{\omega^m}{\omega} \sin \omega. \end{aligned} \quad (3)$$

- b) Prove that R is a rotation (orthogonal with $\det R = 1$) and does not rotate the axis of rotation.

Hint: Consider $\text{tr } x \cdot \sigma x \cdot \sigma$ and $\text{tr } x' \cdot \sigma x' \cdot \sigma$.

Solution: $\text{tr } x \cdot \sigma x \cdot \sigma = x^i x^j \text{tr } \sigma_i \sigma_j = 2\vec{x}^2$ and $\text{tr } x' \cdot \sigma x' \cdot \sigma = \text{tr}(u^{-1}x \cdot \sigma u)(u^{-1}x \cdot \sigma u) = \text{tr } x \cdot \sigma x \cdot \sigma = 2(R^i_j x^j)^2$, hence $\vec{x}^2 = (\vec{x}')^2$. So R preserves lengths. Then $x^T R^T R x = x^T x$, and by considering $x + \alpha y$, we find $x^T R^T R y = x^T y$. Then $R^T R = I$, thus R is orthogonal. Then $\det R = \pm 1$, but at $\omega = 0 \Rightarrow u = 1$ we get $R = I$, hence $\det R = 1$. Thus R lies in $SO(3)$. And $R^k_l \omega^l = \omega^k$: R does not rotate the axis of rotation $\vec{\omega}$.

- c) Consider the homomorphism $\phi(u) = R$ in a). What is the kernel?

Solution: The kernel of $\phi(u) = R$ are those group elements of $SU(2)$ for which $\phi(u) = I$. So $u^{-1}\sigma_k u = \sigma_k$. We already calculated $u^{-1}\sigma_k u$; it gave $R^k_l \sigma_l$. So we must find those $\vec{\omega}$ for which $R^k_l = \delta^k_l$. That is the case if $\cos \omega = 1$ (in which case $\sin \omega = 0$). Then $\omega = 0$ or $\omega = 2\pi$, which gives $u = I$ and $u = -I$. So the kernel is $(I, -I)$ which forms the group Z_2 .

Problem 3. The Lorentz group $L = O(3, 1)$.

The group $O(3, 1)$ preserves $x^\mu \eta_{\mu\nu} x^\nu$ with $\eta_{\mu\nu} = (-1, +1, +1, +1)$, hence $x'^\mu = L^\mu{}_\nu x^\nu$ with $L^T \eta L = \eta$.

a) Show that $\det L = \pm 1$.

Solution: If $L^T \eta L = \eta$, then $(\det L)^2 = 1$, so $\det L = \pm 1$.

b) Show that $L^0{}_0$ is either ≥ 1 or ≤ -1 .

Solution: From $L^T \eta L = \eta$ or $(L^T)^\mu{}_\nu \eta_{\nu\rho} L^\rho{}_\sigma = \eta_{\mu\sigma}$ we get for $\mu = \sigma = 0$

$$\begin{aligned} L^\nu{}_0 \eta_{\nu\rho} L^\rho{}_0 &= \eta_{00} = -1 \\ -(L^0{}_0)^2 + \sum_{j=1}^3 (L^j{}_0)^2 &= -1. \end{aligned} \quad (4)$$

Hence $(L^0{}_0)^2 = 1 + \sum (L^j{}_0)^2 \geq 1$. Then $|L^0{}_0| \geq 1$, or $L^0{}_0 \geq 1$ or $L^0{}_0 \leq -1$.

c) So $O(3, 1)$ contains 4 disconnected components. Write

$$O(3, 1) = SO(3, 1)^\uparrow \cup \Lambda_P SO(3, 1)^\uparrow \cup \Lambda_T SO(3, 1)^\uparrow \cup \Lambda_{PT} SO(3, 1)^\uparrow. \quad (5)$$

Find diagonal matrices for Λ_P , Λ_T and Λ_{PT} .

Solution: The four sets of group elements with $\det L = \pm 1$ and $L^0{}_0 \geq 1$ or ≤ -1 cannot be connected by a continuous curve. (One cannot continuously transform from $\det L = +1$ to $\det L = -1$, or from $L^0{}_0 \geq 1$ to $L^0{}_0 \leq -1$). Any group element g in one of these set can be written as a coset representative g_R times a group element with $\det L = 1$ and $L^0{}_0 \geq 1$, because $g = g_R(g_R^{-1}g)$ and $g_R^{-1}g$ has $\det(g_R^{-1}g) = 1$ and $L^0{}_0(g_R^{-1}g) \geq 1$. The latter follows from $L^{-1} = \eta L^T \eta$ which implies $(L^{-1})^0{}_0 = L^0{}_0$ and $(L^{-1})^0{}_j = -L^j{}_0$. Then $L(g_R^{-1}g)^0{}_0$ is equal to $L(g_R^{-1})^0{}_0 L(g)^0{}_0 + L(g_R^{-1})^0{}_j L(g)^j{}_0 = L(g_R)^0{}_0 L(g)^0{}_0 - L(g_R)^0{}_j L(g)^j{}_0$ and the triangle inequality proves that the first term dominates. As coset representatives we choose

$$\Lambda_P = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}, \quad \Lambda_T = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \quad \Lambda_{PT} = \begin{pmatrix} -1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}. \quad (6)$$

d) Consider the homomorphism $\phi : L \rightarrow (\det L, \text{sign } L^0{}_0)$. Which group V do the four group elements $(\det L, \Lambda^0{}_0) = (\pm 1, \pm 1)$ form?

Solution: The set of 4 group elements $e = (1, 1)$, $a = (1, -1)$, $b = (-1, 1)$ and $c = (-1, -1)$ forms the Klein group $V = (e, a, b, c)$ because $ab = c$, $ac = b$, $bc = a$, $a^2 = b^2 = c^2 = 1$ and it is abelian.

- e) Now show that $SO(3,1)^\uparrow$ is the kernel of the homomorphism $O(3,1) \rightarrow V$. Is $SO(3,1)^\uparrow$ a normal subgroup of $O(3,1)$?

Solution: The map ϕ which maps any group element L of $O(3,1)$ to a group element $(\det L, \text{sign } L^0_0)$ is a homomorphism because

$$\det L_1 L_2 = \det L_1 \det L_2 \text{ and } \text{sign } (L_1 L_2)^0_0 = \text{sign } (L_1)^0_0 \text{sign } (L_2)^0_0. \quad (7)$$

The latter we proved in c). The kernel consists of those L with $\det L = +1$ and $L^0_0 \geq 1$. That is $SO(3,1)^\uparrow$. Since $SO(3,1)^\uparrow$ is the kernel, it is a normal subgroup of $O(3,1)$.

Problem 4.

We want to prove that $Sl(2, \mathbb{C})$ is the covering group of $SO(3,1)^\uparrow$. Consider the expression $Ax^\mu \sigma_\mu A^\dagger$ with A in $Sl(2, \mathbb{C})$ and $\sigma_\mu = (-I, \sigma_1, \sigma_2, \sigma_3)$. (Note $A^\dagger$, not A^{-1} this time.)

- a) Show that this expression is equal to $x'^\mu \sigma_\mu$, where $x'^\mu = L^\mu_\nu x^\nu$ with **real** L^μ_ν .

Hint: Can one always write a complex 2×2 matrix as $x'^\mu \sigma_\mu$?

Solution: $Ax^\mu \sigma_\mu A^\dagger$ is a 2×2 matrix. Any 2×2 matrix can be expanded on σ^k, I . So $Ax^\mu \sigma_\mu A^\dagger = x'^\mu \sigma_\mu$. However, $Ax^\mu \sigma_\mu A^\dagger$ is linear in x^ν so x'^μ is linear in x^ν . Then $x'^\mu = L^\mu_\nu x^\nu$. Since $Ax^\mu \sigma_\mu A^\dagger$ is hermitian, x'^μ is real. So L^μ_ν is real.

- b) Show that $\det x^\mu \sigma_\mu = \det x'^\mu \sigma_\mu$, and then show that $L^\mu_\nu \in O(3,1)$.

Solution: $\det Ax^\mu \sigma_\mu A^\dagger = \det x^\mu \sigma_\mu$ because $\det A = \det A^\dagger = 1$. Then

$$\det x^\mu \sigma_\mu = \det \begin{pmatrix} -x^0 + x^3 & x^1 - ix^2 \\ x^1 + ix^2 & -x^0 - x^3 \end{pmatrix} = (x^0)^2 - (x^3)^2 - (x^1)^2 - (x^2)^2 \equiv -x^\mu \eta_{\mu\nu} x^\nu. \quad (8)$$

Similarly $\det x'^\mu \sigma_\mu = -x'^\mu \eta_{\mu\nu} x'^\nu = -(L^\mu_\rho x^\rho) \eta_{\mu\nu} (L^\nu_\sigma x^\sigma) = -x^\rho (L^T)_\rho^\mu \eta_{\mu\nu} L^\nu_\sigma x^\sigma = -x^T L^T \eta L x$. So $x^T \eta x = x^T L^T \eta L x$. Replacing x by $x + \alpha y$ we obtain (done in class for $SO(N)$)

$$x^T \eta y = x^T L^T \eta L y \quad \Rightarrow \quad L^T \eta L = \eta. \quad (9)$$

- c) Now show that $L^\mu_\nu \in SO(3,1)^\uparrow$. We denote these group elements by $\hat{L}^\mu_\nu$.

Solution: From (9) it follows that L lies in the group $O(3,1)$. But for $A \rightarrow I$ we get $x'^\mu = x^\mu$, so $L = I$. So L is connected to the unit I . So $\det L = +1$ and $L^0_0 \geq 1$. Thus L lies in $SO(3,1)^\uparrow$.

- d) Show that the map $\phi(A) = \hat{L}$ is a homomorphism. What is the kernel of ϕ ?

Solution: $BA\sigma_\mu x^\mu A^\dagger B^\dagger = B(A\sigma_\mu x^\mu A^\dagger)B^\dagger = B(x'^\mu \sigma_\mu)B^{-1} = L_B^\mu_\nu x'^\nu \sigma_\mu = L_B^\mu_\nu L_A^\nu_\rho x^\rho \sigma_\mu$ since $x'^\nu = L_A^\nu_\rho x^\rho$. On the other hand $BA\sigma_\mu x^\mu A^\dagger B^\dagger = (BA)\sigma_\mu x^\mu (BA)^\dagger = x''^\mu \sigma_\mu$ with

$x'^\mu = L_{BA}{}^\mu{}_\nu x^\nu$. So $L_B L_A = L_{BA}$. Then $\phi(BA) = L_{BA} = L_B L_A = \phi(B)\phi(A)$. so $\phi(A) = \hat{L}$ is a homomorphism. The kernel of ϕ are all matrices A such that $Ax^\mu \sigma_\mu A^\dagger = x^\mu \sigma_\mu$. Two solutions are obvious: $A = I$ and $A = -I$. To show that there are no more we set $A = aI + b^k \sigma_k$ with complex a and b^k satisfying $a^2 - \sum_{k=0}^3 (b^k)^2 = 1$. Then we must solve

$$(aI + b^k \sigma_k) \sigma_\mu (a^* I + (b^l)^* \sigma_l) = \sigma_\mu. \quad (10)$$

For $\sigma_\mu = I$ we get $|a|^2 + \sum |b^k|^2 = 1$ and $\epsilon_{klm} b^k (b^l)^* = 0$. Clearly $a^2 = |a|^2 = 1$, so $a = \pm 1$.

e) Show that $Sl(2, \mathbb{C})/Z_2 \simeq SO(3, 1)^\dagger$.

Solution: Since $Z_2 = (I, -I)$ is the kernel of the homomorphism of $SU(2)$ into $SO(3)$ we have $Sl(2, \mathbb{C})/Z_2 = SO(3, 1)$ if we can prove that the homomorphism is not only “into” but also “onto”. Given an A in $Sl(2, \mathbb{C})$ we always get an L in $SO(3, 1)$. All L can be obtained from the pure boosts and pure rotations. The matrices $e^{\vec{\omega} \cdot \vec{\sigma}}$ give all boosts and the matrices $A = e^{i\vec{\omega} \cdot \vec{\sigma}}$ give all rotations. Hence we get the whole $SO(3, 1)$.

Bonus Problem.

Show that $Sl(2, \mathbb{C})$ is simply connected.

a) First assume that $A \in Sl(2, \mathbb{C})$ has the polar decomposition $A = HU$ where $U \in SU(2)$ and H is a 2×2 matrix which is hermitian, has $\det H = 1$, and $\text{tr } H > 0$. Show that the manifold of H is a connected component of a hyperboloid. Then use this to show that $Sl(2, \mathbb{C})$ is simply connected.

Solution: If $A = HU$ with A in $Sl(2, \mathbb{C})$ and $U \in SU(2)$ and H hermitian, then $\det H = 1$. We get $AA^\dagger = HH$. Since $AA^\dagger$ is hermitian, it can be decomposed by a unitary matrix V , $AA^\dagger = V^{-1} \tilde{D} V$ with $\tilde{D}$ diagonal. Since $AA^\dagger > 0$ (meaning $x^\dagger AA^\dagger x > 0$ for all x) $\tilde{D}$ has diagonal entries > 0 . Then we can take the square root D of $\tilde{D}$ by just taking the square root of the diagonal entries. So $\tilde{D} = D^2$ and $D > 0$. Then $AA^\dagger = (V^{-1} D V)(V^{-1} D V)$ so we define H by $V^{-1} D V = H$. So we constructed H from A . This H indeed satisfies $\det H = 1$, $H^\dagger = H$, and $\text{tr } H > 0$ (because $D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ with $\lambda_1 > 0$ and $\lambda_2 > 0$). Finally, to satisfy $A = HU$, we define U by $U = H^{-1}A$. We must prove that U lies in $SU(2)$: $\det U = 1$ and $UU^\dagger = I$. We already showed that $\det H = \det A = 1$, so we only must show that $UU^\dagger = I$

$$\begin{aligned} UU^\dagger &= H^{-1} AA^\dagger H^{-1} = (V^{-1} D^{-1} V) AA^\dagger (V^{-1} D^{-1} V) \\ &= (V^{-1} D^{-1} V) [(V^{-1} D V)(V^{-1} D V)] (V^{-1} D^{-1} V) \\ &= V^{-1} D^{-1} D D D^{-1} V = I. \end{aligned} \quad (11)$$

b) Show that the matrices H do not form a group (if group multiplication is matrix multiplication).

Solution: The product of two hermitian matrices is in general not hermitian.

c) Now prove the polar decomposition $A = HU$.

Hint: First diagonalize $AA^\dagger$ by a unitary matrix V such that $VAA^\dagger V^{-1} = D^2$. Let D be the positive square root of D^2 , so $\text{tr } D > 0$. Then define $H = V^{-1}DV$.

Solution: See a).

d) Show that $U = H^{-1}A$ lies in $SU(2)$.

Solution: See a).

e) We showed above that $Sl(2, \mathbb{C})$ is simply connected. Is $Gl(2, \mathbb{C})$ also simply connected?

Solution: Consider the map $\phi(g) = \det g$ for g in $Gl(2, \mathbb{C})$. The whole complex plane except the origin is covered by $\det g$. Consider a closed curve around the origin in the complex plane. It corresponds to a closed curve in $Gl(2, \mathbb{C})$. The contraction of this curve yields a point where $\det g = 0$, but $\det g \neq 0$ for groups. Thus $Gl(2, \mathbb{C})$ is not simply connected.

A proof that follows more what we did for $SU(2)/Z_2$ is as follows: Begin with $Gl(N, \mathbb{C})$ and write g in $Gl(N, \mathbb{C})$ as $g_0 z$ where g_0 lies in $Sl(N, \mathbb{C})$ and $z \neq 0$ is a complex number. The decomposition of g into g_0 and z is not unique: one gets the same g if one replaces g_0 by $g_0 e^{2\pi i k/N}$ (which still lies in $Sl(N, \mathbb{C})$) and replaces z by $z e^{-2\pi i k/N}$ which is still a complex number different from 0. The manifold of these points is $C \setminus 0$, the complex plane with the origin removed. Hence we divide by the group elements $e^{2\pi i k/N}$ which form the group Z_N

$$Gl(N, \mathbb{C}) = \frac{Sl(N, \mathbb{C}) \times C \setminus 0}{Z_N}. \quad (12)$$

Now $N = 2$. The group $Sl(2, \mathbb{C})$ is simply connected and the discrete group Z_2 is also simply connected, but $C \setminus 0$ is infinitely connected. Hence $Gl(2, \mathbb{C})$ is not simply connected.